Miguel Chavez Nava

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EDUCATION

Emory University **Atlanta, GA**
Bachelor of Science in Computer Science; Bachelor of Arts in Film & Media Studies May 2027
GPA: 3.9 / 4.0

TECHNICAL SKILLS

Core Competencies: Data Structures & Algorithms, Artificial Intelligence, Debugging & Testing, Data Visualization, Version Control
Programming Languages: Python, Java, JavaScript, TypeScript, C/C++, SQL, HTML/CSS, Assembly
Developer Tools: Git, VS Code, PyTorch, Sk-Learn, Numpy, Pandas, React, Firebase, Google Cloud Services, Unity, D3, UNIX

PROGRAMMING PROJECTS

Computing in Cardiology Challenge | Python, *https://miguelchaveznava.github.io/challenge.html* Feb 2026

- Engineered a logistic regression mortality prediction model trained on 12,000 patient records, achieving an AUROC of 0.85 using L2 regularization, outperforming baseline models by prioritizing clinical signal extraction.
- Constructed a 50+ dimension feature set by computing mean and standard deviation across time-series patient data, enabling richer model inputs and improving predictive accuracy.
- Leveraged scikit-learn to evaluate multiple model architectures, accelerating the iteration cycle and identifying optimal configurations across 3+ candidate designs.

Raging Gambler | C++, Unity, Git, *https://miguelchaveznava.github.io/gambler.html* Feb 2025 - May 2025

- Co-developed a top-down arcade-style action game in Unity with a team of 6, achieving a fully playable product across 40 iterative builds and was experienced by 100+ players.
- Architected 5 dynamic difficulty-scaling algorithms and adjusted enemy behavior/game state in real time based on player progression, enhancing replayability and player engagement.
- Championed version control practices using GitHub, resolving 15+ merge conflicts, and organizing collaborative development workflows across the full team and supervising code integrity throughout the project lifecycle.

PPM Image Transformation | C, UNIX, *https://miguelchaveznava.github.io/transform.html* Jan 2025

- Developed a command-line image processing tool in C capable of executing 8 distinct PPM transformation algorithms, delivering a flexible utility for low-level image manipulation.
- Engineered 10 configurable terminal arguments for transformation selection and I/O handling, giving users precise control while sustaining a sub-1-second processing runtime.
- Applied memory-efficient structs and optimized algorithms to handle image data at scale, demonstrating strong systems programming fundamentals in a constrained UNIX environment.

WORK EXPERIENCE

Emory Department of Dining **Atlanta, GA**
Student Assistant Aug 2023 - Present

- Deliver dining services and resolve vendor inquiries for 50+ students weekly, maintaining a consistent and welcoming experience supporting daily campus operations.
- Streamline coordination of campus dining events and resources by managing 20+ monthly calls, ensuring clear communication between departments and improving event execution efficiency.
- Facilitate 10+ vendor transactions and provide customer support at the weekly Emory Farmer's Market, contributing to a smooth marketplace experience for hundreds of campus visitors.

Carl's Jr Restaurants **Las Vegas, NV**
Cashier May 2023 - Aug 2025

- Processed 100+ customer transactions daily with speed and accuracy, sustaining a less than 2% error rate and contributing to a smooth front-of-house operation.
- Delivered attentive customer service during high-volume rushes, solving 10+ order discrepancies per shift and ensuring a positive dining experience and repeat visits.
- Collaborated with a team of 15+ staff to maintain efficient workflows during peak hours, reducing average customer wait time by an estimated 20% through proactive communication and task coordination.

ADDITIONAL INFORMATION

Languages: English, Spanish
Interests: Films (Nolan, Scorsese, Kubrick), Video Games, Music
Engagement: Emory Computer Science Club (Vice President of Communications), Emory Cinematheque